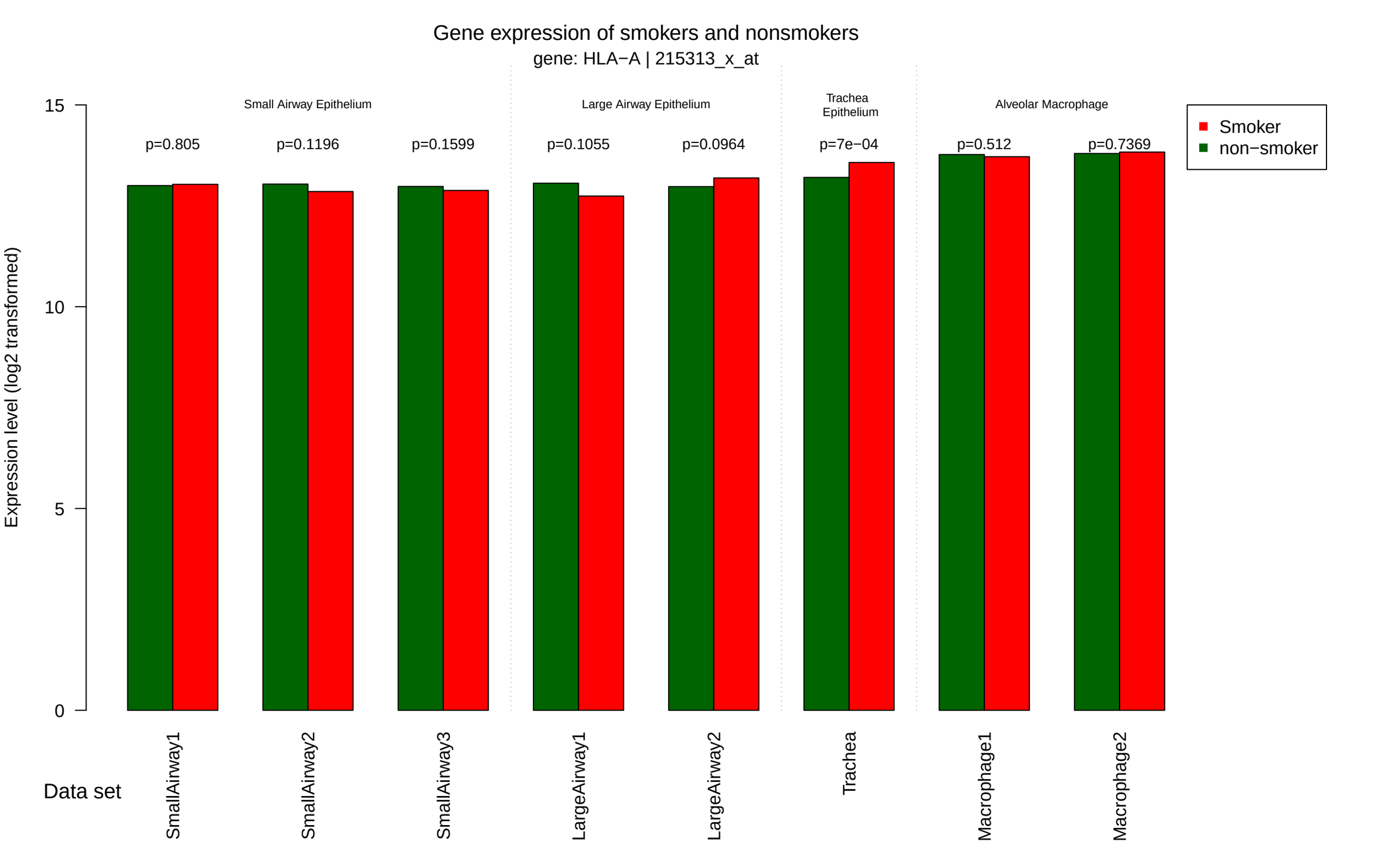
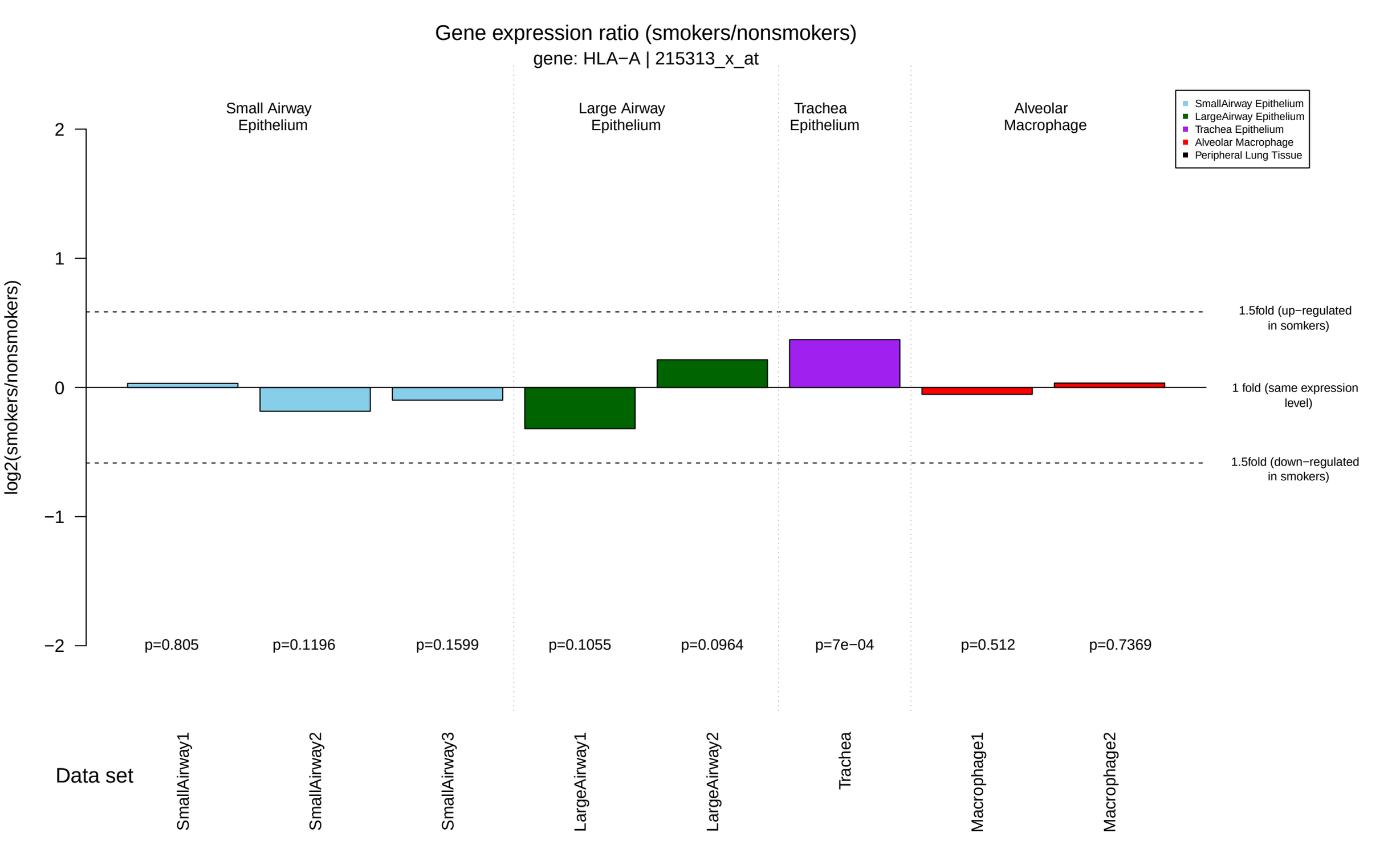
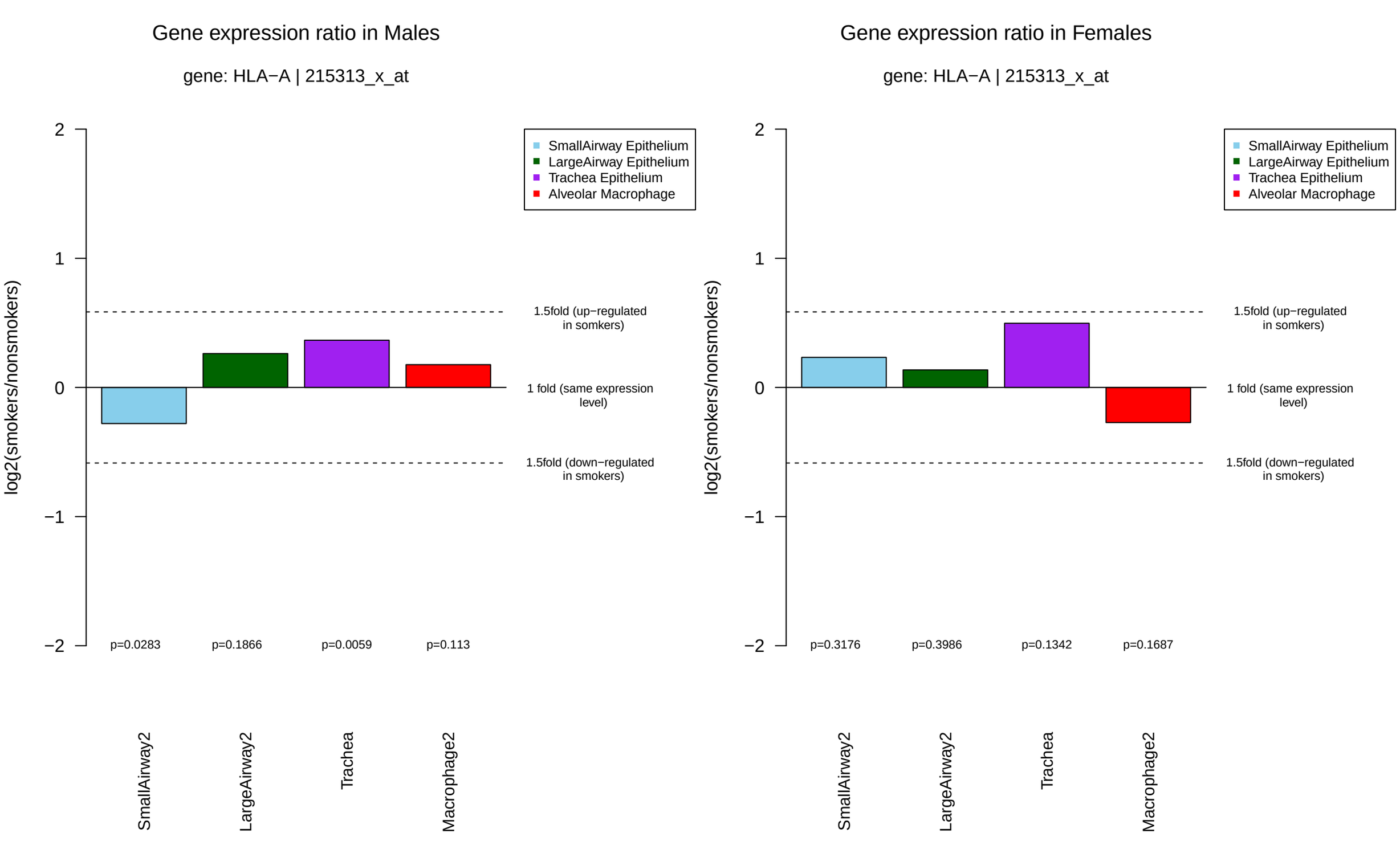
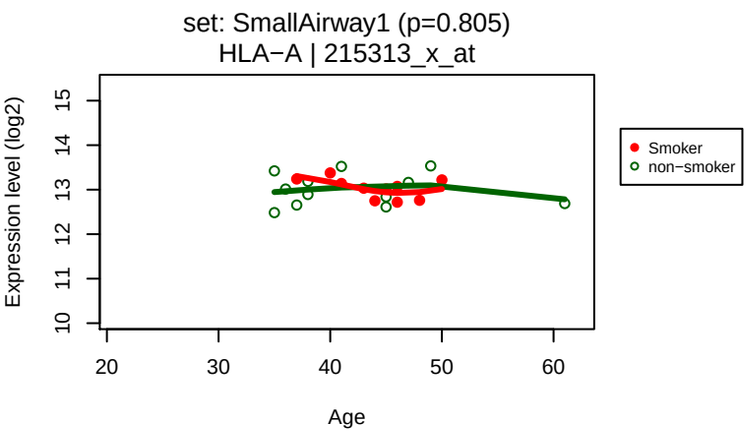
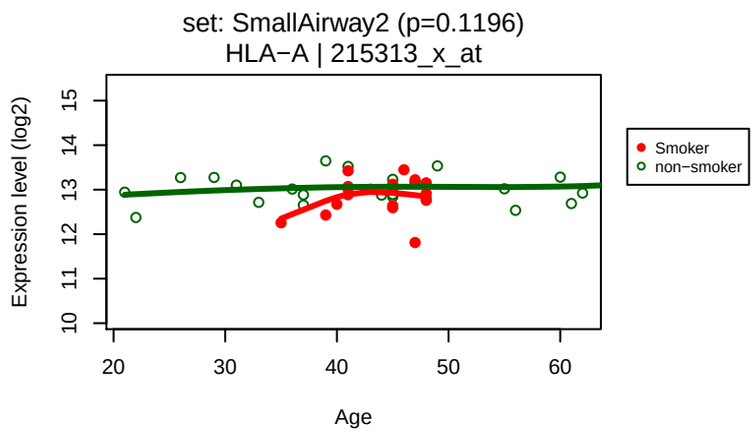
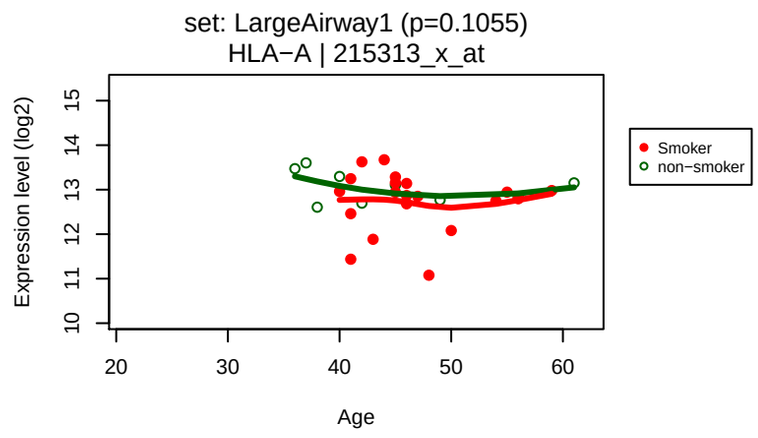

Gene expression of smokers and nonsmokers
gene: HLA−A | 215313_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.805
p=0.1196
p=0.1599
p=0.1055
p=0.0964
p=7e−04
p=0.512
p=0.7369
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

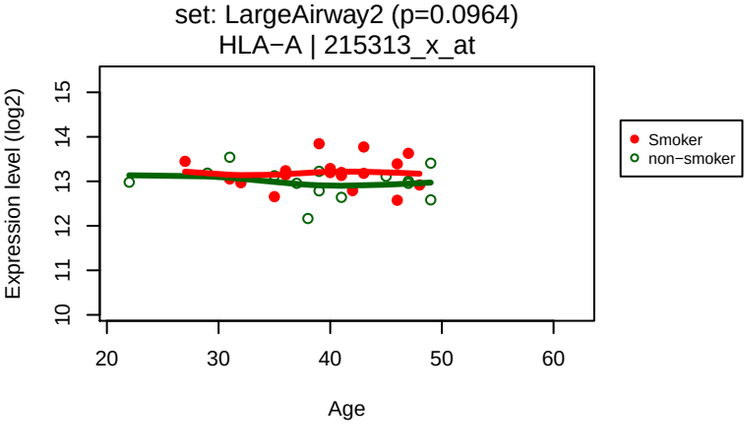

Gene expression ratio (smokers/nonsmokers)
gene: HLA−A | 215313_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.805
p=0.1196
p=0.1599
p=0.1055
p=0.0964
p=7e−04
p=0.512
p=0.7369
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

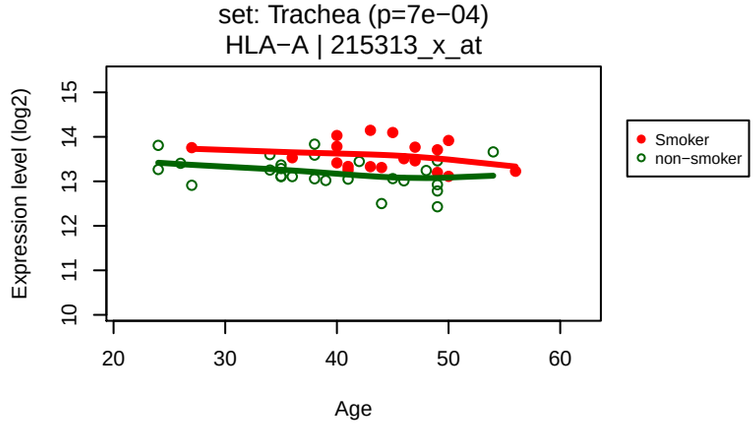

Gene expression ratio in Males
gene: HLA−A | 215313_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0283
p=0.1866
p=0.0059
p=0.113
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−A | 215313_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3176
p=0.3986
p=0.1342
p=0.1687
SmallAirway2
LargeAirway2
Trachea
Macrophage2

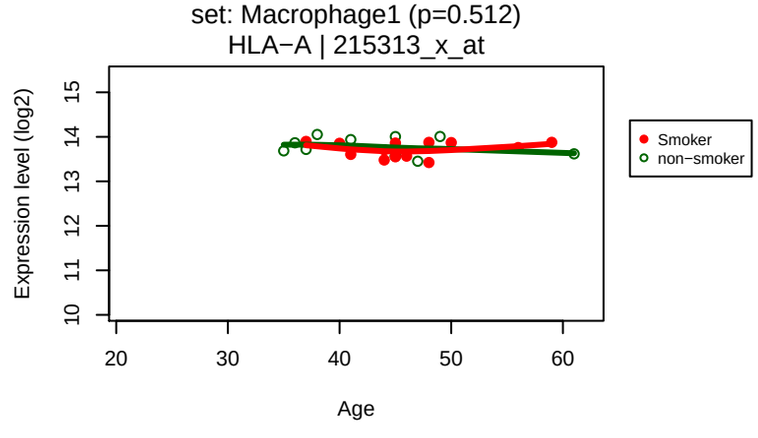

set: SmallAirway1 (p=0.805)
HLA−A | 215313_x_at
Expression level (log2)
Age
Smoker
non−smoker

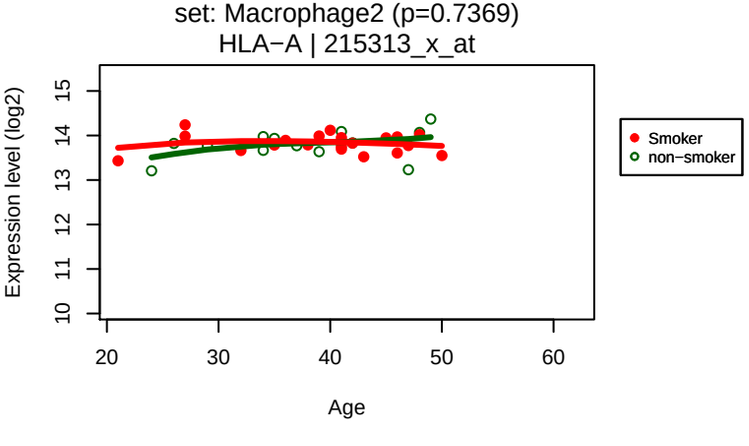

set: SmallAirway2 (p=0.1196)
HLA−A | 215313_x_at
Expression level (log2)
Age
Smoker
non−smoker

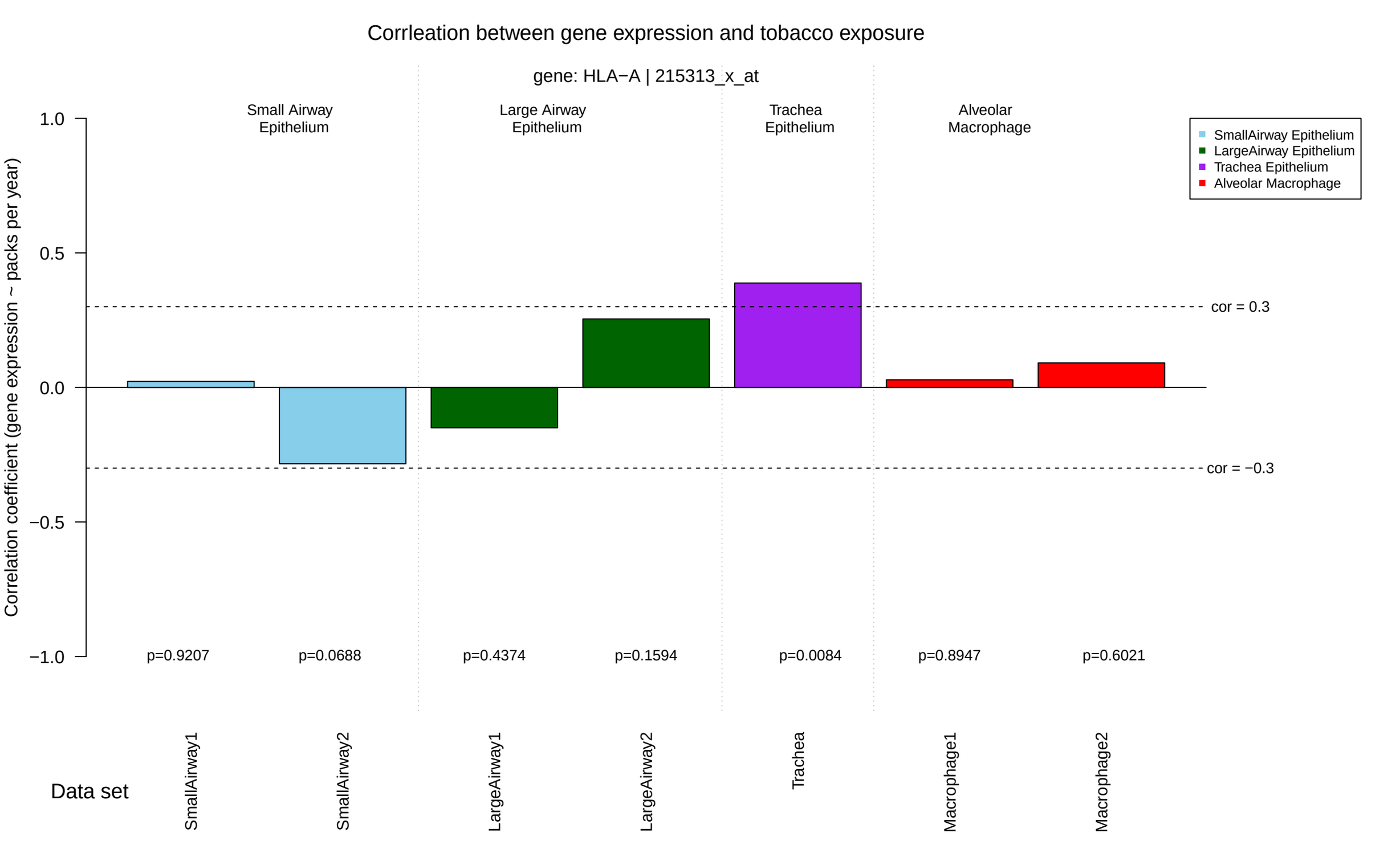

set: LargeAirway1 (p=0.1055)
HLA−A | 215313_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0964)
HLA−A | 215313_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=7e−04)
HLA−A | 215313_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.512)
HLA−A | 215313_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7369)
HLA−A | 215313_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−A | 215313_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9207
p=0.0688
p=0.4374
p=0.1594
p=0.0084
p=0.8947
p=0.6021
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2